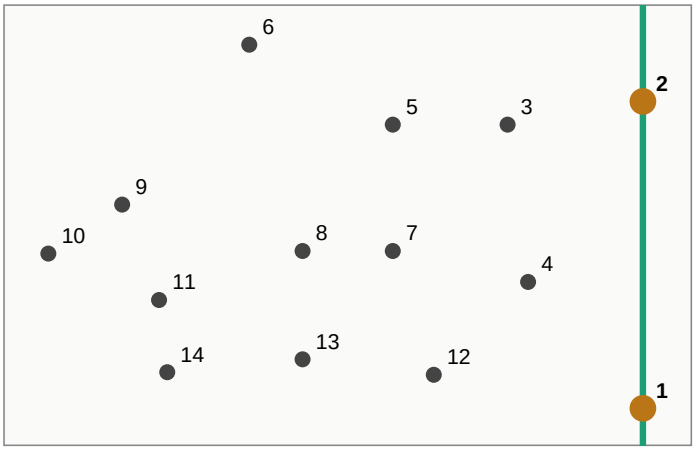
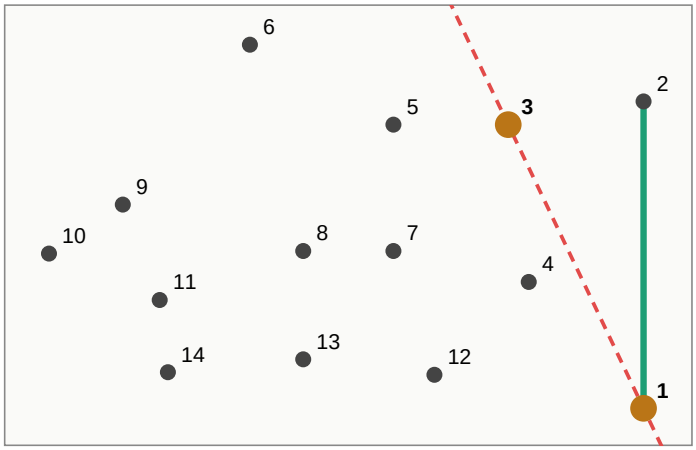
Convex hull: 1. Brute force

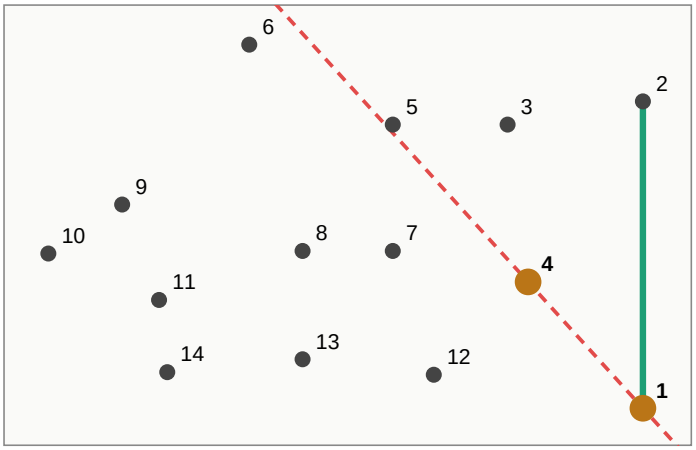
Green = hull edge / stack Amber = being tested Red dashed = rejected / right turn Grey = helper lines (points read visually from Figure 2)



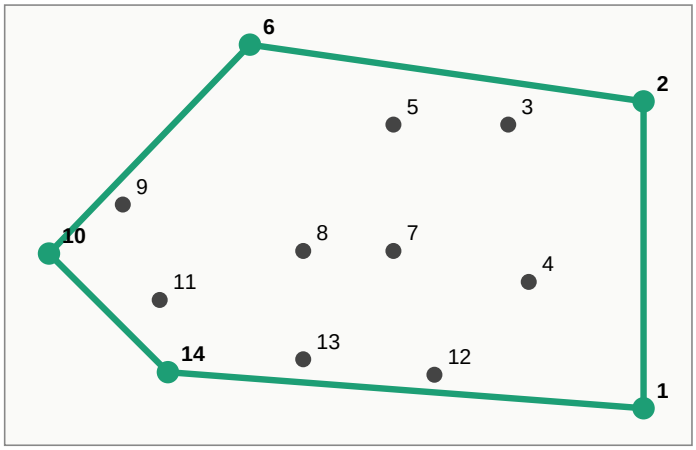
Step 1 / 8
Pair (1,2): draw the line through them. All other points lie on its left, so 1-2 is a hull edge.



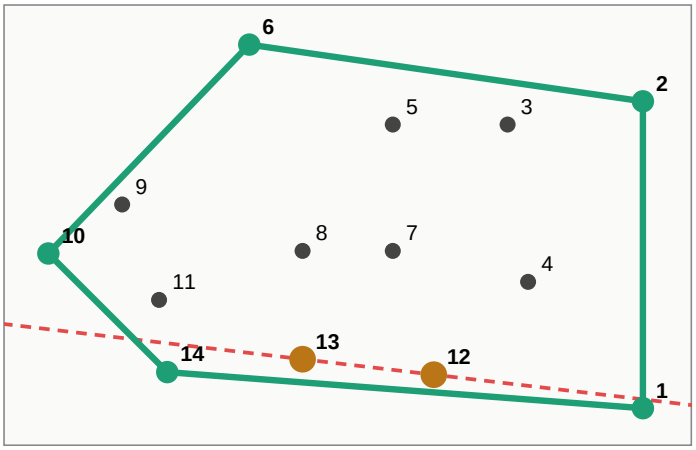
Step 2 / 8
Pair (1,3): points fall on both sides of the line. Rejected.



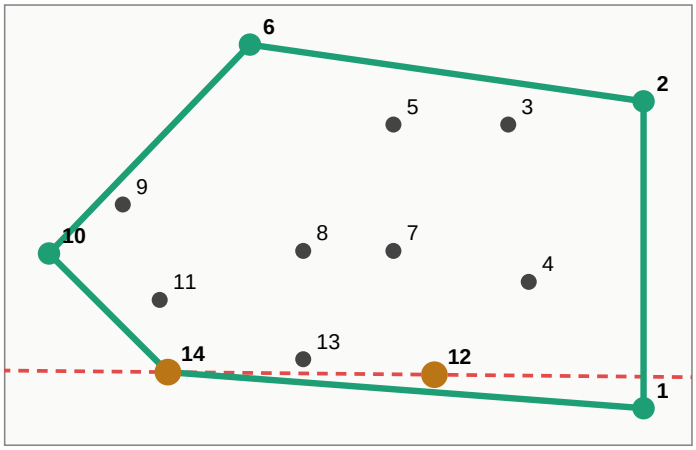
Step 3 / 8
Pair (1,4): points on both sides again. Rejected.



Step 4 / 8
Pairs 4 to 88 skipped. Testing all pairs finds exactly 5 valid edges: 1-2, 2-6, 6-10, 10-14, 14-1.



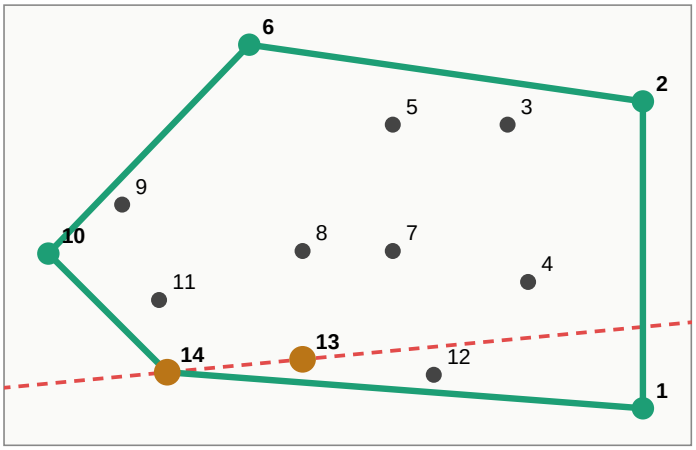
Step 5 / 8
Third-last pair (12,13): points 10 and 14 are on opposite sides. Rejected.



Step 6 / 8
Second-last pair (12,14): point 13 above, point 1 below. Rejected.

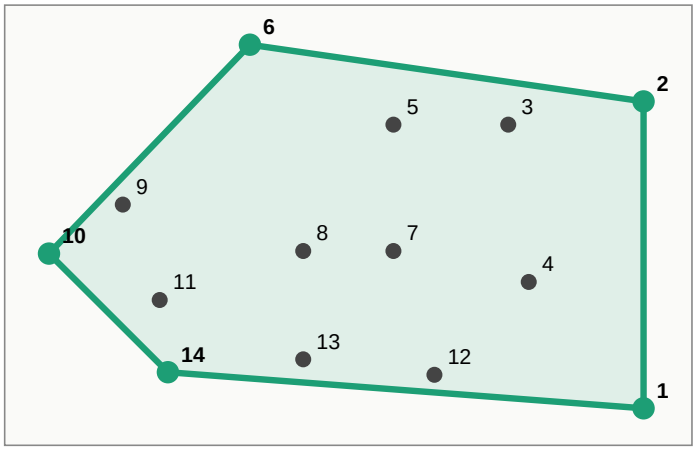
Convex hull: 1. Brute force

Green = hull edge / stack Amber = being tested Red dashed = rejected / right turn Grey = helper lines (points read visually from Figure 2)



Step 7 / 8

Last pair (13,14): point 1 below, point 2 above. Rejected. All 91 pairs checked.

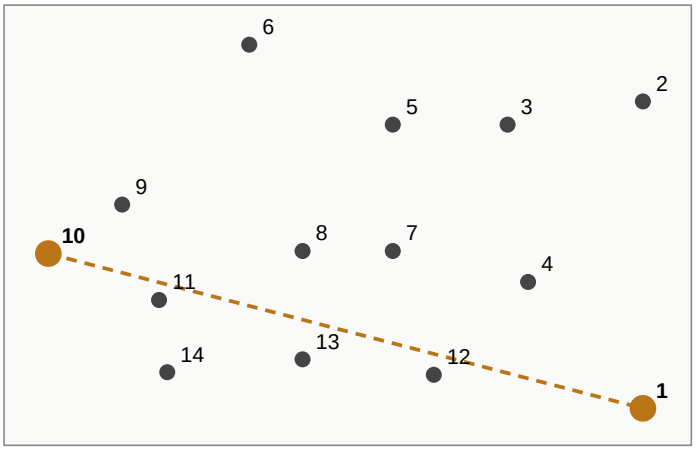


Step 8 / 8

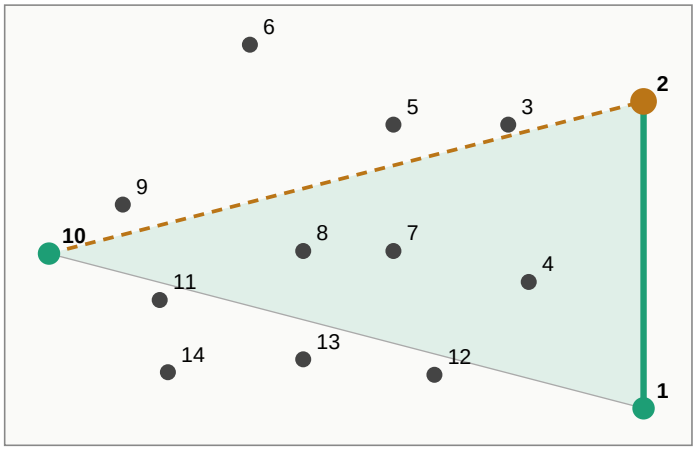
Result: hull 10 -> 14 -> 1 -> 2 -> 6.

Convex hull: 2. Quick hull

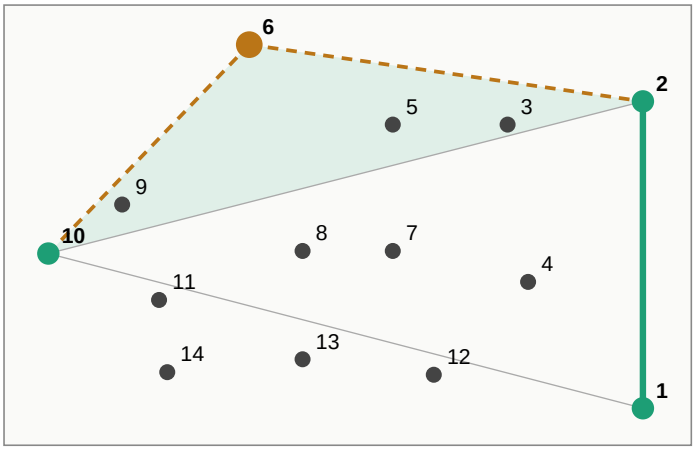
Green = hull edge / stack Amber = being tested Red dashed = rejected / right turn Grey = helper lines (points read visually from Figure 2)



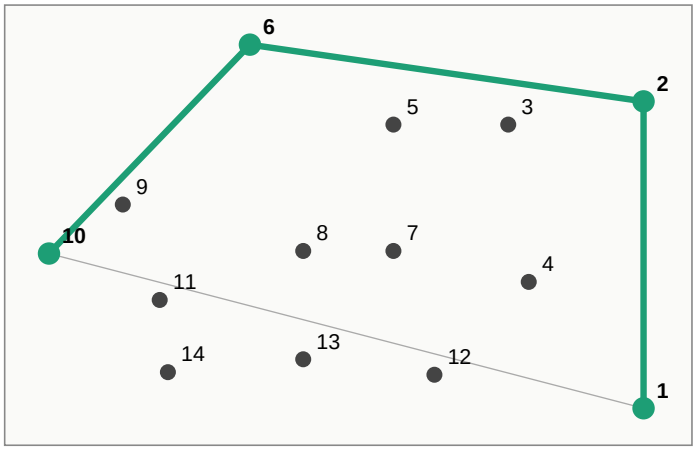
Step 1 / 6
Find leftmost (10) and rightmost (1) points. Draw line 10-1; it splits the points into upper and lower sets.



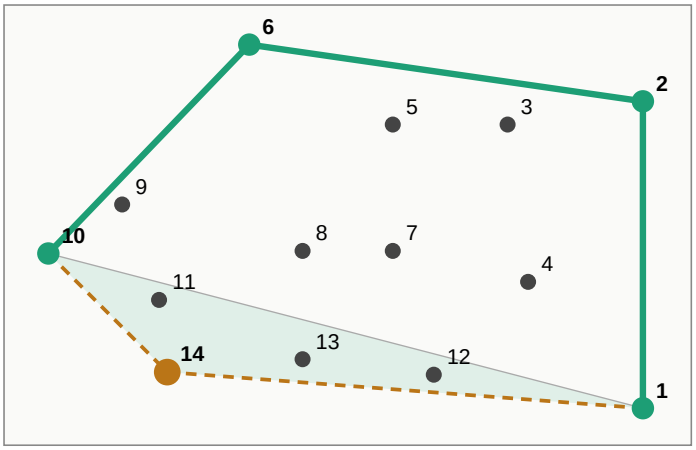
Step 2 / 6
Upper set: farthest point from 10-1 is 2. Form triangle 10-2-1; points inside are discarded.



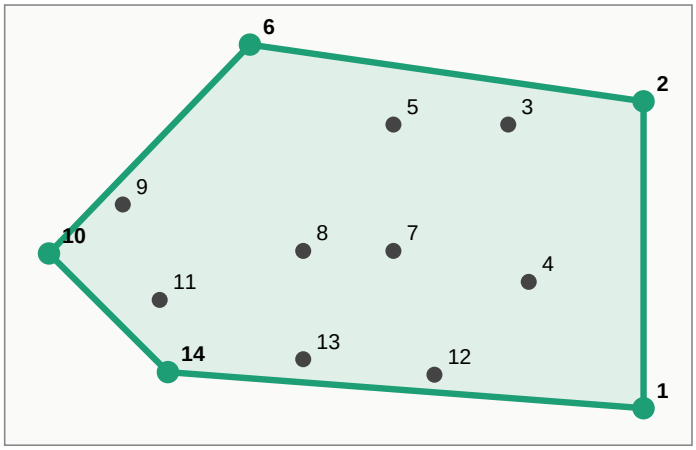
Step 3 / 6
Outside edge 10-2: points 9, 6, 5, 3. The farthest is 6. Form triangle 10-6-2.



Step 4 / 6
Edges 10-6 and 6-2 have no points outside. Upper hull done: 10, 6, 2, 1.



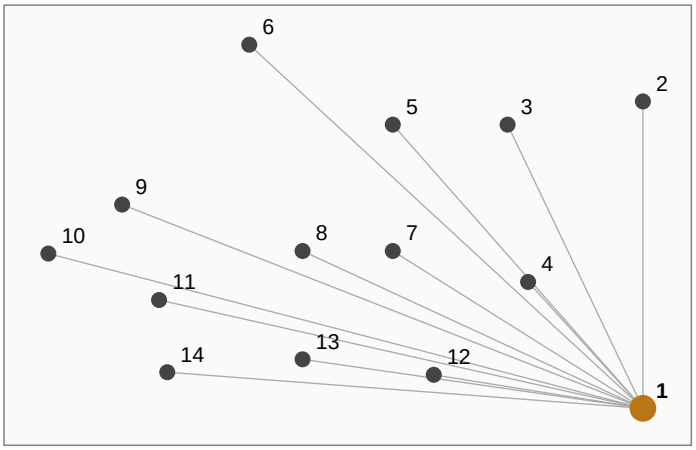
Step 5 / 6
Lower set: 11, 13, 12, 14. Farthest from 10-1 is 14. Form triangle 10-14-1.



Step 6 / 6
Edges 10-14 and 14-1 have nothing outside. Recursion stops. Hull: 10, 6, 2, 1, 14.

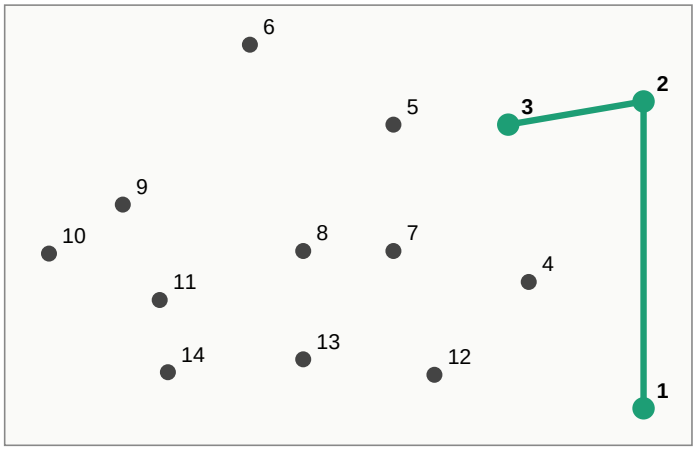
Convex hull: 3. Graham scan

Green = hull edge / stack Amber = being tested Red dashed = rejected / right turn Grey = helper lines (points read visually from Figure 2)



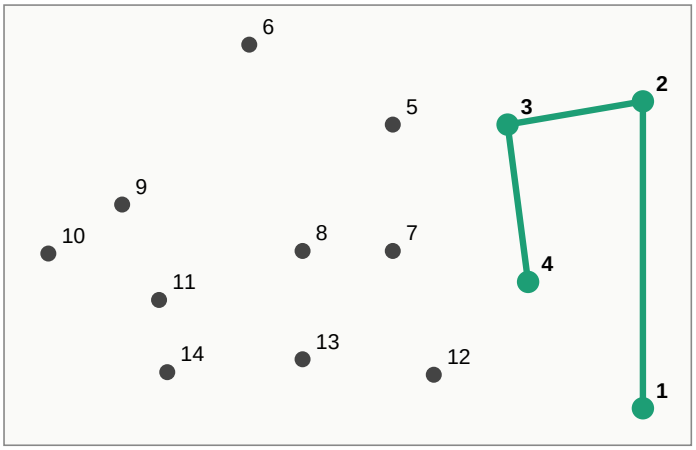
Step 1 / 9

P = point 1 is the lowest point. Sort all others by polar angle around P: 2, 3, 4, ..., 14.



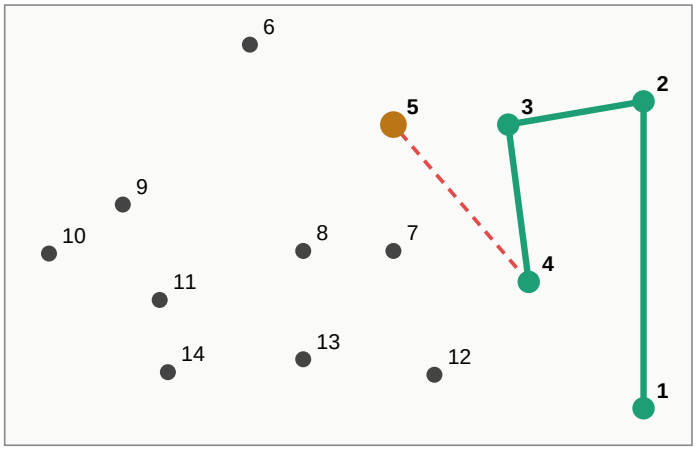
Step 2 / 9

Push 1, 2, 3. Turn 1->2->3 is a left turn, keep all three.



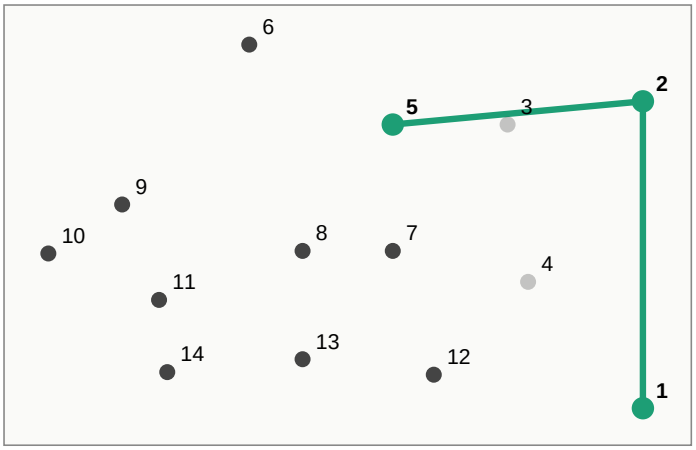
Step 3 / 9

Push 4. Turn 2->3->4 is a left turn, keep. Stack: 1, 2, 3, 4.



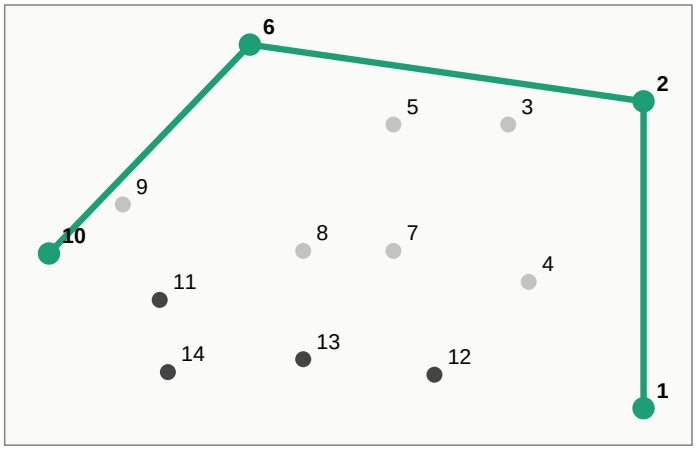
Step 4 / 9

Consider 5. Turn 3->4->5 is a right turn (red): pop 4. Then 2->3->5 is also right: pop 3.



Step 5 / 9

After popping 4 and 3, push 5. Stack: 1, 2, 5.

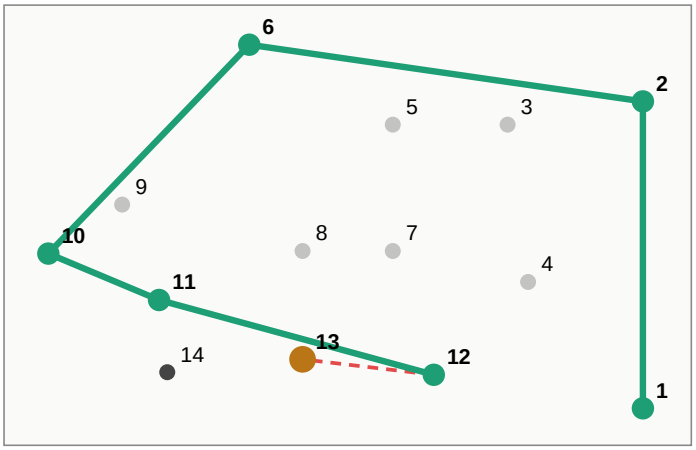


Step 6 / 9

Points 6 to 10 processed the same way; 5, 7, 8, 9 are popped by right turns. Stack: 1, 2, 6, 10.

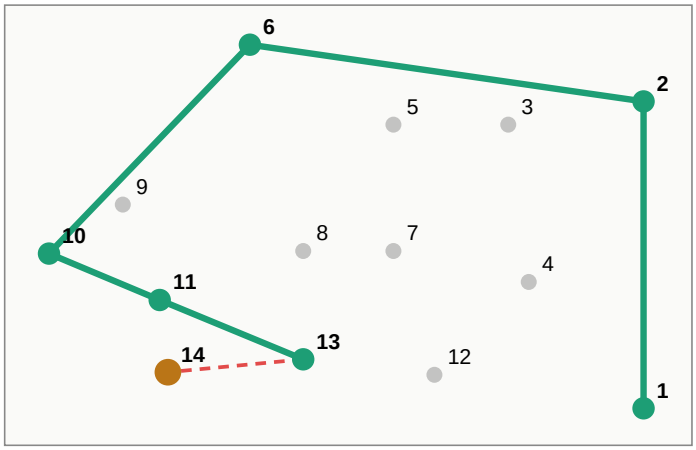
Convex hull: 3. Graham scan

Green = hull edge / stack Amber = being tested Red dashed = rejected / right turn Grey = helper lines (points read visually from Figure 2)



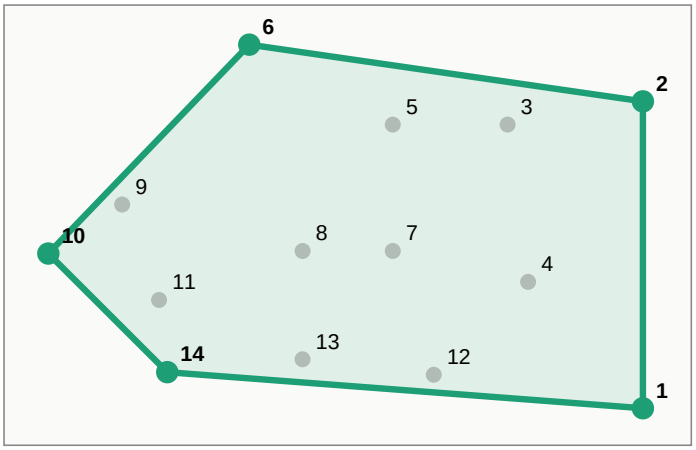
Step 7 / 9

Push 11, then 12 (left turns). Consider 13: turn 11->12->13 is right, so 12 is popped.



Step 8 / 9

Pop 12, push 13. Consider 14: 11->13->14 is right, pop 13; 10->11->14 is right, pop 11.

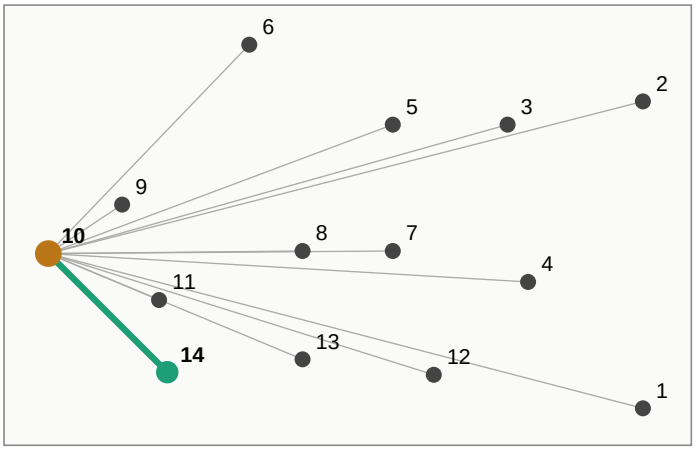


Step 9 / 9

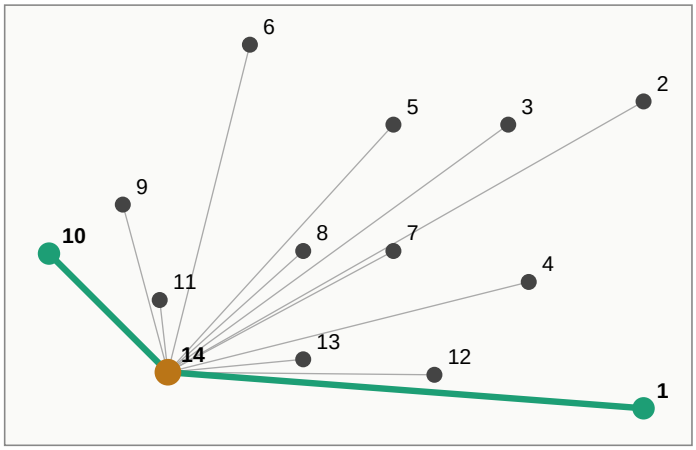
Push 14 and close back to P. Final stack: 1, 2, 6, 10, 14.

Convex hull: 4. Jarvis march (gift wrapping)

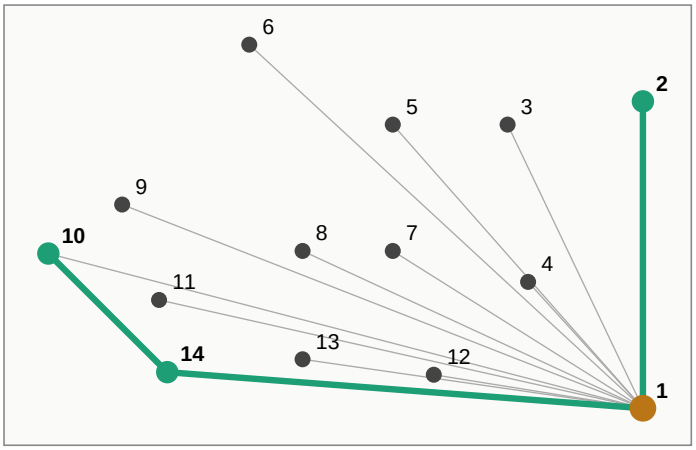
Green = hull edge / stack Amber = being tested Red dashed = rejected / right turn Grey = helper lines (points read visually from Figure 2)



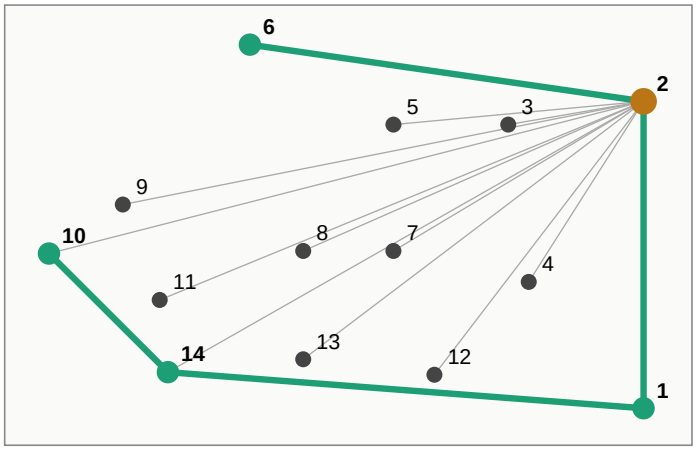
Step 1 / 5
Start at leftmost point 10. Test directions to all others; pick the one with all points on its left: 14. Edge 10->14.



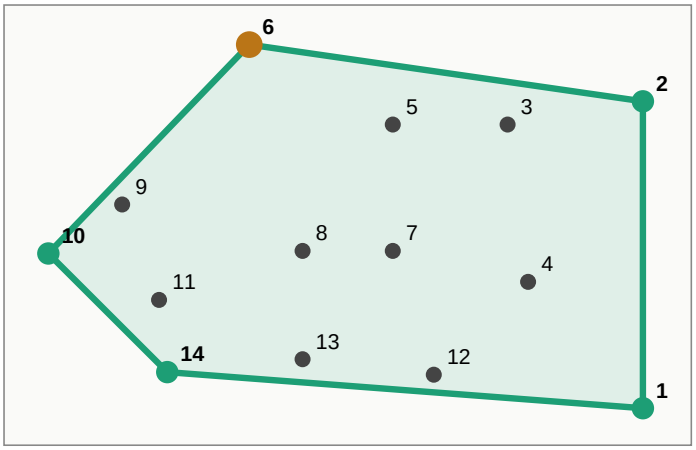
Step 2 / 5
From 14, the most counter-clockwise point is 1. Edge 14->1.



Step 3 / 5
From 1, the next point is 2 (straight up). Edge 1->2.



Step 4 / 5
From 2, wrap around to 6. Edge 2->6.



Step 5 / 5
From 6, wrap to 10 (start). The march stops. Hull: 10 -> 14 -> 1 -> 2 -> 6 (h = 5).